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A Training and Transfer Details

This appendix provides the full training and transfer setup underlying the main results. We repeat the core setup from the main text for completeness and add the optimization and implementation details omitted there for space.

A.1 Backbones and Transfer Setup

We instantiate StreamGuard across multiple backbone families and sizes. Our primary scaling experiments use Llama3 backbones at 1B, 3B, and 8B. To study portability beyond a single tokenizer family, we additionally evaluate transfer to Gemma and Qwen backbones at small model scales.

Because our prefix-level supervision is primarily constructed with the Llama3 tokenizer, we study transfer by applying these Llama3-based targets to Gemma and Qwen backbones. In the native setting, prefix targets can be consumed directly as tokenizer-aligned token labels. In the transfer setting, where tokenizer coordinates do not match, the same supervision is reused in text space by pairing decoded prefixes with future-risk targets. This contrasts with boundary-based streaming supervision, where unsafe onset indices are tied to tokenizer-specific coordinates and are therefore not directly portable across architectures. We compare this transferred setting against a native same-tokenizer setting.

A.2 Reference Training and Target Construction

Training and target construction use examples from WILDGUARDTRAIN, AEGIS2.0, TOXICCHAT, and BEAVERTAILS. Unless otherwise noted, all main results use a fixed reference StreamGuard target-construction setup. This includes mean reduction, a four-model rollout pool, and a fixed supervision schedule.

For the reference rollout pool, we sample continuations from Llama 3 (Grattafiori et al., 2024) 8B and 70B, Qwen2.5 (Qwen et al., 2025) 7B and 72B, drawing four rollouts per model at temperature 0.7. We score the rollouts using Qwen3Guard-8B-strict (Zhao et al., 2025). Rollout expansion is only required for unsafe prompts, since unsafe responses arise only when the corresponding prompts are unsafe.

Because rollout-based target construction is expensive for long responses, our reference training setup uses a budgeted supervision schedule on longer-response datasets such as WILDGUARDTRAIN, applying prefix supervision densely over an initial segment and more sparsely thereafter. In the main paper, rollout-source composition, reduction, and supervision density are treated as practical implementation choices and analyzed in Section 5.

A.3 Implementation Details

Training Configuration. We train all models with a maximum sequence length of 8192 tokens and an effective global batch size of 128, using gradient accumulation as needed. We determine the learning rate through a grid search over $\{2e\text{-}6, 5e\text{-}6, 8e\text{-}6, 1e\text{-}5\}$, selecting $8e$-6 for 8B models and $1e$-5 for smaller models. Optimization uses AdamW with default hyperparameters except for the learning rate, together with a `StepLR` scheduler with step size 1 epoch and decay factor $\gamma = 0.85$. All models are trained in bfloat16 with full-parameter fine-tuning for up to three epochs, using gradient clipping with a maximum norm of 1.0 and no warmup. We merge all training datasets into a single training corpus. Checkpoints are selected based on F1 on the AEGIS2.0 official validation set (Ghosh et al., 2025). All experiments use custom PyTorch (Paszke et al., 2019) training code with FSDP (Zhao et al., 2023). Unless otherwise noted, each model variant is trained with three random seeds, and we report the mean and standard deviation across runs.

Architecture and Inference. We attach a linear classification head directly to the final hidden state of every token in the sequence. This enables the model to produce a continuous risk estimate $V_\theta(x, y_{\le t})$ at every step of the generation without altering the input structure.

Model	Streaming	ToxiC	OAIMod	Aegis	Aegis2	SSTest	HarmB	WildG	Avg
LlamaGuard3-8B	no	53.8	79.5	71.5	76.4	99.5	99.0	76.4	79.4
LlamaGuard4-12B	no	51.3	73.5	67.8	70.6	98.0	97.2	73.0	75.9
WildGuard-7B	no	70.8	72.1	89.4	80.7	99.5	98.9	88.9	85.8
ShieldGemma-9B (Zeng et al., 2024)	no	69.4	82.1	70.3	72.5	83.7	60.6	54.2	70.4
ShieldGemma-27B	no	72.9	80.5	69.0	71.6	84.4	57.3	54.3	70.0
NemoGuard-8B	no	75.6	81.0	81.4	86.8	98.5	75.2	81.6	82.9
PolyGuard-Qwen-7B (Kumar et al., 2025)	no	71.5	74.1	90.3	86.3	100.0	98.7	88.1	87.0
Qwen3Guard-0.6B-Gen-strict	no	65.1	66.5	90.8	85.0	99.0	98.7	87.7	84.7
Qwen3Guard-0.6B-Gen-loose	no	77.7	77.6	76.9	83.3	95.8	96.1	85.1	84.6
Qwen3Guard-4B-Gen-strict	no	69.5	68.3	90.8	85.8	99.5	100.0	85.6	85.6
Qwen3Guard-4B-Gen-loose	no	82.8	80.7	76.3	82.1	97.4	99.2	85.1	86.2
Qwen3Guard-8B-Gen-strict	no	68.9	68.8	91.4	86.1	99.5	100.0	88.9	86.2
Qwen3Guard-8B-Gen-loose	no	82.8	81.3	76.0	82.5	97.4	98.5	85.6	86.3
Qwen3Guard-Stream-0.6B-strict	yes	72.0	68.3	85.2	84.9	98.0	97.2	87.1	84.7
Qwen3Guard-Stream-0.6B-loose	yes	75.5	76.0	77.7	81.7	96.9	96.8	86.0	84.4
Qwen3Guard-Stream-4B-strict	yes	73.0	70.0	85.9	86.6	**99.5**	**100.0**	88.6	86.2
Qwen3Guard-Stream-4B-loose	yes	81.7	81.2	75.5	80.2	98.5	98.9	85.3	85.9
Qwen3Guard-Stream-8B-strict	yes	75.3	74.0	85.7	86.1	99.0	99.4	87.5	86.7
Qwen3Guard-Stream-8B-loose	yes	**80.1**	**80.3**	75.5	80.8	98.5	98.7	84.4	85.5
Llama3-StreamGuard-1B **(ours)**	yes	74.9± 0.3	71.5± 0.2	**89.0**± 0.3	87.7± 0.2	98.6± 0.2	94.8± 2.9	88.6± 0.2	86.4
Llama3-StreamGuard-3B **(ours)**	yes	77.7± 0.5	74.4± 0.3	87.1± 0.4	87.8± 0.1	99.3± 0.2	99.2± 0.5	89.0± 0.2	87.8
Llama3-StreamGuard-8B **(ours)**	yes	77.4± 0.5	75.0± 0.4	88.5± 0.2	**87.9**± 0.1	**99.5**± 0.0	99.7± 0.1	**89.5**± 0.2	**88.2**

Table 6: Input-moderation F1; Avg denotes the macro-average. For prior systems, we report published results. We report mean ± std over three seeds for StreamGuard models. The *Streaming* column indicates whether a model supports token-level streaming moderation.

At test time, a prefix is classified as unsafe when its predicted score exceeds a threshold, which defaults to 0.5, with the final trigger determined by the debouncing rule described in Section 3.1.

B Full Standard Moderation Results

For space, the main text reports all streaming baselines together with a reduced set of offline references. This appendix restores the full baseline panel for the standard input- and output-moderation comparisons. These expanded tables preserve the same overall conclusions as the main text: StreamGuard remains competitive with strong offline post-hoc moderators while improving over prior streaming baselines in the setting most relevant to this paper.

B.1 Input Moderation

Table 6 reports the full input-moderation comparison, including all offline and streaming baselines. The main text keeps the discussion focused on the strongest and most relevant comparisons for the streaming setting, while this appendix restores the complete model set for transparency. The full table confirms that the main-text pattern is not an artifact of baseline selection: StreamGuard remains strong on the macro-average and particularly competitive on newer safety benchmarks such as AEGIS2.0 and WILDGUARDTEST, while the looser Qwen variants remain relatively stronger on TOXICCHAT and OPENAIMODERATION, consistent with the policy-mismatch.

B.2 Output Moderation

Table 7 reports the full output-moderation comparison under the same streaming response-level protocol used in the main text. As with input moderation, the main paper emphasizes the comparisons most central to our claim—namely, how StreamGuard compares to prior streaming guardrails and to strong offline post-hoc references—while this appendix restores the full baseline set. The expanded table confirms the same qualitative conclusion as the main text: StreamGuard remains among the strongest models in the causal streaming setting, and its gains over prior streaming baselines are not explained by selective baseline reporting.